The file `tulm.fdd` for use with LaTeX 2ε.*

LaTeX Project

2021/07/23

This file is maintained by the LaTeX Project team.
Bug reports can be opened (category `latex`) at
https://latex-project.org/bugs.html.

1 TUlmr

```
⟨*TUlmr⟩
\DeclareFontFamily{TU}{lmr}{}
\DeclareFontShape{TU}{lmr}{m}{n}%
  {<-5.5>    \UnicodeFontFile{lmroman5-regular}{\UnicodeFontTeXLigatures}
   <5.5-6.5> \UnicodeFontFile{lmroman6-regular}{\UnicodeFontTeXLigatures}
   <6.5-7.5> \UnicodeFontFile{lmroman7-regular}{\UnicodeFontTeXLigatures}
   <7.5-8.5> \UnicodeFontFile{lmroman8-regular}{\UnicodeFontTeXLigatures}
   <8.5-9.5> \UnicodeFontFile{lmroman9-regular}{\UnicodeFontTeXLigatures}
   <9.5-11>  \UnicodeFontFile{lmroman10-regular}{\UnicodeFontTeXLigatures}
   <11-15>   \UnicodeFontFile{lmroman12-regular}{\UnicodeFontTeXLigatures}
   <15-> \UnicodeFontFile{lmroman17-regular}{\UnicodeFontTeXLigatures}
   }{}
\DeclareFontShape{TU}{lmr}{m}{sl}%
  {<-8.5>    \UnicodeFontFile{lmromanslant8-regular}{\UnicodeFontTeXLigatures}
   <8.5-9.5> \UnicodeFontFile{lmromanslant9-regular}{\UnicodeFontTeXLigatures}
   <9.5-11>  \UnicodeFontFile{lmromanslant10-regular}{\UnicodeFontTeXLigatures}
   <11-15>   \UnicodeFontFile{lmromanslant12-regular}{\UnicodeFontTeXLigatures}
   <15-> \UnicodeFontFile{lmromanslant17-regular}{\UnicodeFontTeXLigatures}
   }{}
\DeclareFontShape{TU}{lmr}{m}{it}%
  {<-7.5>    \UnicodeFontFile{lmroman7-italic}{\UnicodeFontTeXLigatures}
   <7.5-8.5> \UnicodeFontFile{lmroman8-italic}{\UnicodeFontTeXLigatures}
   <8.5-9.5> \UnicodeFontFile{lmroman9-italic}{\UnicodeFontTeXLigatures}
   <9.5-11>  \UnicodeFontFile{lmroman10-italic}{\UnicodeFontTeXLigatures}
   <11->   \UnicodeFontFile{lmroman12-italic}{\UnicodeFontTeXLigatures}
   }{}
\DeclareFontShape{TU}{lmr}{m}{sc}%
  {<-> \UnicodeFontFile{lmromancaps10-regular}{\UnicodeFontTeXLigatures}}{}
\DeclareFontShape{TU}{lmr}{m}{ui}%
  {<-> \UnicodeFontFile{lmromanunsl10-regular}{\UnicodeFontTeXLigatures}}{}
\DeclareFontShape{TU}{lmr}{m}{scsl}%
  {<-> \UnicodeFontFile{lmromancaps10-oblique}{\UnicodeFontTeXLigatures}}{}
```

*This file has version number v1.8a, dated 2021/07/23.

```
\DeclareFontShape{TU}{lmr}{b}{n}
  {<-> \UnicodeFontFile{lmromandemi10-regular}{\UnicodeFontTeXLigatures}}{}
\DeclareFontShape{TU}{lmr}{b}{sl}
  {<-> \UnicodeFontFile{lmromandemi10-oblique}{\UnicodeFontTeXLigatures}}{}
\DeclareFontShape{TU}{lmr}{bx}{n}
  {<-5.5>   \UnicodeFontFile{lmroman5-bold}{\UnicodeFontTeXLigatures}
   <5.5-6.5> \UnicodeFontFile{lmroman6-bold}{\UnicodeFontTeXLigatures}
   <6.5-7.5> \UnicodeFontFile{lmroman7-bold}{\UnicodeFontTeXLigatures}
   <7.5-8.5> \UnicodeFontFile{lmroman8-bold}{\UnicodeFontTeXLigatures}
   <8.5-9.5> \UnicodeFontFile{lmroman9-bold}{\UnicodeFontTeXLigatures}
   <9.5-11>  \UnicodeFontFile{lmroman10-bold}{\UnicodeFontTeXLigatures}
   <11->   \UnicodeFontFile{lmroman12-bold}{\UnicodeFontTeXLigatures}
   }{}
\DeclareFontShape{TU}{lmr}{bx}{it}
  {<-> \UnicodeFontFile{lmroman10-bolditalic}{\UnicodeFontTeXLigatures}}{}
\DeclareFontShape{TU}{lmr}{bx}{sl}
  {<-> \UnicodeFontFile{lmromanslant10-bold}{\UnicodeFontTeXLigatures}}{}
\DeclareFontShape{TU}{lmr}{b}{it}
  {<->sub * lmr/b/sl}{}
⟨/TUlmr⟩
```

2 TUlmss

```
⟨*TUlmss⟩
\DeclareFontFamily{TU}{lmss}{}
\DeclareFontShape{TU}{lmss}{m}{n}
  {<-8.5> \UnicodeFontFile{lmsans8-regular}{\UnicodeFontTeXLigatures}
   <8.5-9.5> \UnicodeFontFile{lmsans9-regular}{\UnicodeFontTeXLigatures}
   <9.5-11>  \UnicodeFontFile{lmsans10-regular}{\UnicodeFontTeXLigatures}
   <11-15.5> \UnicodeFontFile{lmsans12-regular}{\UnicodeFontTeXLigatures}
   <15.5-> \UnicodeFontFile{lmsans17-regular}{\UnicodeFontTeXLigatures}
   }{}
\DeclareFontShape{TU}{lmss}{m}{it}
   {<->ssub*lmss/m/sl}{}
\DeclareFontShape{TU}{lmss}{m}{sl}
  {<-8.5> \UnicodeFontFile{lmsans8-oblique}{\UnicodeFontTeXLigatures}
   <8.5-9.5> \UnicodeFontFile{lmsans9-oblique}{\UnicodeFontTeXLigatures}
   <9.5-11>  \UnicodeFontFile{lmsans10-oblique}{\UnicodeFontTeXLigatures}
   <11-15.5> \UnicodeFontFile{lmsans12-oblique}{\UnicodeFontTeXLigatures}
   <15.5-> \UnicodeFontFile{lmsans17-oblique}{\UnicodeFontTeXLigatures}
   }{}
\DeclareFontShape{TU}{lmss}{m}{sc}
  {<->sub*lmr/m/sc}{}
\DeclareFontShape{TU}{lmss}{b}{n}
  {<->ssub * lmss/bx/n}{}
\DeclareFontShape{TU}{lmss}{b}{sl}
  {<->ssub * lmss/bx/sl}{}
\DeclareFontShape{TU}{lmss}{b}{it}
  {<->ssub * lmss/bx/it}{}
\DeclareFontShape{TU}{lmss}{sbc}{n}
  {<-> \UnicodeFontFile{lmsansdemicond10-regular}{\UnicodeFontTeXLigatures}}{}
\DeclareFontShape{TU}{lmss}{sbc}{sl}
  {<-> \UnicodeFontFile{lmsansdemicond10-oblique}{\UnicodeFontTeXLigatures}}{}
\DeclareFontShape{TU}{lmss}{sbc}{it}
```

```
  {<->ssub*lmss/sbc/sl}{}
\DeclareFontShape{TU}{lmss}{bx}{n}
  {<-> \UnicodeFontFile{lmsans10-bold}{\UnicodeFontTeXLigatures}}{}
\DeclareFontShape{TU}{lmss}{bx}{sl}
  {<-> \UnicodeFontFile{lmsans10-boldoblique}{\UnicodeFontTeXLigatures}}{}
\DeclareFontShape{TU}{lmss}{bx}{it}
  {<->ssub*lmss/bx/sl}{}
```
⟨/TUlmss⟩

3 TUlmssq

```
⟨*TUlmssq⟩
\DeclareFontFamily{TU}{lmssq}{}
\DeclareFontShape{TU}{lmssq}{m}{n}
  {<-> \UnicodeFontFile{lmsansquot8-regular}{\UnicodeFontTeXLigatures}}{}
\DeclareFontShape{TU}{lmssq}{m}{sl}
  {<-> \UnicodeFontFile{lmsansquot8-oblique}{\UnicodeFontTeXLigatures}}{}
\DeclareFontShape{TU}{lmssq}{bx}{n}
  {<-> \UnicodeFontFile{lmsansquot8-bold}{\UnicodeFontTeXLigatures}}{}
\DeclareFontShape{TU}{lmssq}{bx}{sl}
  {<-> \UnicodeFontFile{lmsansquot8-boldoblique}{\UnicodeFontTeXLigatures}}{}
\DeclareFontShape{TU}{lmssq}{m}{it}
  {<->ssub*lmssq/m/sl}{}
\DeclareFontShape{TU}{lmssq}{bx}{it}
  {<->ssub*lmssq/bx/sl}{}
\DeclareFontShape{TU}{lmssq}{b}{n}
  {<->ssub * lmssq/bx/n}{}
\DeclareFontShape{TU}{lmssq}{b}{sl}
  {<->ssub * lmssq/bx/sl}{}
\DeclareFontShape{TU}{lmssq}{b}{it}
  {<->ssub * lmssq/bx/it}{}
```
⟨/TUlmssq⟩

4 TUlmtt

```
⟨*TUlmtt⟩
\DeclareFontFamily{TU}{lmtt}{%
\hyphenchar \font\m@ne
\fontdimen3\font\z@
\fontdimen4\font\z@
}
\ifx\lmtt@use@light@as@normal\@empty
\DeclareFontShape{TU}{lmtt}{sb}{n}
  {<-8.5>   \UnicodeFontFile{lmmono8-regular}{}
   <8.5-9.5> \UnicodeFontFile{lmmono9-regular}{}
   <9.5-11> \UnicodeFontFile{lmmono10-regular}{}
   <11->  \UnicodeFontFile{lmmono12-regular}{}
   }{}
\DeclareFontShape{TU}{lmtt}{sb}{it}
  {<-> \UnicodeFontFile{lmmono10-italic}{}}{}
\DeclareFontShape{TU}{lmtt}{sb}{sl}
  {<-> \UnicodeFontFile{lmmonoslant10-regular}{}}{}
\DeclareFontShape{TU}{lmtt}{sb}{sc}
  {<-> \UnicodeFontFile{lmmonocaps10-regular}{}}{}
```

```
\DeclareFontShape{TU}{lmtt}{sb}{scsl}
  {<-> \UnicodeFontFile{lmmonocaps10-oblique}{}}{}
\DeclareFontShape{TU}{lmtt}{m}{n}
  {<-> \UnicodeFontFile{lmmonolt10-regular}{}}{}
\DeclareFontShape{TU}{lmtt}{m}{it}
  {<->sub*lmtt/l/sl}{}
\DeclareFontShape{TU}{lmtt}{m}{sl}
  {<-> \UnicodeFontFile{lmmonolt10-oblique}{}}{}
\DeclareFontShape{TU}{lmtt}{c}{n}
  {<-> \UnicodeFontFile{lmmonoltcond10-regular}{}}{}
\DeclareFontShape{TU}{lmtt}{c}{it}
  {<->sub*lmtt/lc/sl}{}
\DeclareFontShape{TU}{lmtt}{c}{sl}
  {<-> \UnicodeFontFile{lmmonoltcond10-oblique}{}}{}
\else
\DeclareFontShape{TU}{lmtt}{m}{n}
  {<-8.5>   \UnicodeFontFile{lmmono8-regular}{}
  <8.5-9.5> \UnicodeFontFile{lmmono9-regular}{}
   <9.5-11> \UnicodeFontFile{lmmono10-regular}{}
   <11->  \UnicodeFontFile{lmmono12-regular}{}
   }{}
\DeclareFontShape{TU}{lmtt}{m}{it}
  {<-> \UnicodeFontFile{lmmono10-italic}{}}{}
\DeclareFontShape{TU}{lmtt}{m}{sl}
  {<-> \UnicodeFontFile{lmmonoslant10-regular}{}}{}
\DeclareFontShape{TU}{lmtt}{m}{sc}
  {<-> \UnicodeFontFile{lmmonocaps10-regular}{}}{}
\DeclareFontShape{TU}{lmtt}{m}{scsl}
  {<-> \UnicodeFontFile{lmmonocaps10-oblique}{}}{}
\DeclareFontShape{TU}{lmtt}{l}{n}
  {<-> \UnicodeFontFile{lmmonolt10-regular}{}}{}
\DeclareFontShape{TU}{lmtt}{l}{it}
  {<->sub*lmtt/l/sl}{}
\DeclareFontShape{TU}{lmtt}{l}{sl}
  {<-> \UnicodeFontFile{lmmonolt10-oblique}{}}{}
\DeclareFontShape{TU}{lmtt}{lc}{n}
  {<-> \UnicodeFontFile{lmmonoltcond10-regular}{}}{}
\DeclareFontShape{TU}{lmtt}{lc}{it}
  {<->sub*lmtt/lc/sl}{}
\DeclareFontShape{TU}{lmtt}{lc}{sl}
  {<-> \UnicodeFontFile{lmmonoltcond10-oblique}{}}{}
\fi
\DeclareFontShape{TU}{lmtt}{b}{n}
  {<-> \UnicodeFontFile{lmmonolt10-bold}{}}{}
\DeclareFontShape{TU}{lmtt}{b}{it}
  {<->sub*lmtt/b/sl}{}
\DeclareFontShape{TU}{lmtt}{b}{sl}
{<-> \UnicodeFontFile{lmmonolt10-boldoblique}{}}{}
\DeclareFontShape{TU}{lmtt}{bx}{it}
  {<->sub*lmtt/b/sl}{}
\DeclareFontShape{TU}{lmtt}{bx}{n}
  {<->ssub*lmtt/b/n}{}
\DeclareFontShape{TU}{lmtt}{bx}{sl}
  {<->ssub*lmtt/b/sl}{}
```

```
⟨/TUlmtt⟩
```

5 TUlmvtt

```
⟨*TUlmvtt⟩
\DeclareFontFamily{TU}{lmvtt}{}
\ifx\lmtt@use@light@as@normal\@empty
\DeclareFontShape{TU}{lmvtt}{sb}{n}
  {<->\UnicodeFontFile{lmmonoprop10-regular}{}}{}
\DeclareFontShape{TU}{lmvtt}{sb}{sl}
  {<->\UnicodeFontFile{lmmonoprop10-oblique}{}}{}
\DeclareFontShape{TU}{lmvtt}{sb}{it}
  {<->sub*lmvtt/m/sl}{}
\DeclareFontShape{TU}{lmvtt}{m}{n}
  {<->\UnicodeFontFile{lmmonoproplt10-regular}{}}{}
\DeclareFontShape{TU}{lmvtt}{m}{sl}
  {<->\UnicodeFontFile{lmmonoproplt10-oblique}{}}{}
\DeclareFontShape{TU}{lmvtt}{m}{it}
  {<->sub*lmvtt/l/sl}{}
\else
\DeclareFontShape{TU}{lmvtt}{m}{n}
  {<->\UnicodeFontFile{lmmonoprop10-regular}{}}{}
\DeclareFontShape{TU}{lmvtt}{m}{sl}
  {<->\UnicodeFontFile{lmmonoprop10-oblique}{}}{}
\DeclareFontShape{TU}{lmvtt}{m}{it}
  {<->sub*lmvtt/m/sl}{}
\DeclareFontShape{TU}{lmvtt}{l}{n}
  {<->\UnicodeFontFile{lmmonoproplt10-regular}{}}{}
\DeclareFontShape{TU}{lmvtt}{l}{sl}
  {<->\UnicodeFontFile{lmmonoproplt10-oblique}{}}{}
\DeclareFontShape{TU}{lmvtt}{l}{it}
  {<->sub*lmvtt/l/sl}{}
\fi
\DeclareFontShape{TU}{lmvtt}{b}{n}
  {<->\UnicodeFontFile{lmmonoproplt10-bold}{}}{}
\DeclareFontShape{TU}{lmvtt}{b}{sl}
  {<->\UnicodeFontFile{lmmonoproplt10-boldoblique}{}}{}
\DeclareFontShape{TU}{lmvtt}{b}{it}
  {<->sub*lmvtt/b/sl}{}
\DeclareFontShape{TU}{lmvtt}{bx}{n}
  {<->ssub*lmvtt/b/n}{}
\DeclareFontShape{TU}{lmvtt}{bx}{sl}
  {<->ssub*lmvtt/b/sl}{}
\DeclareFontShape{TU}{lmvtt}{bx}{it}
  {<->sub*lmvtt/b/sl}{}
⟨/TUlmvtt⟩
```

6 TUlmdh

```
⟨*TUlmdh⟩
\DeclareFontFamily{TU}{lmdh}{}%
\DeclareFontShape{TU}{lmdh}{m}{n}
  {<->   \UnicodeFontFile{lmromandunh10-regular}{\UnicodeFontTeXLigatures}
  }{}
```

```
\DeclareFontShape{TU}{lmdh}{m}{sl}
  {<->   \UnicodeFontFile{lmromandunh10-oblique}{\UnicodeFontTeXLigatures}
  }{}
\DeclareFontShape{TU}{lmdh}{m}{it}
  {<->sub*lmdh/m/sl}{}
```
⟨/TUlmdh⟩